

Popcorn Response to Nitrogen Fertilizer – Purdue University Research Overview (2023 – 2025)

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Research and Location Descriptions:



Image 1. Indiana research trial locations for Purdue Univ. popcorn nitrogen trials 2023 – 2025.

Purdue Research Station Locations:

Small plot research trial treatment design (Wanatah, West Lafayette and Butlerville, IN only) included randomized completed block designs with 5 replications. Plot size measured 15 ft wide (6, 30-inch corn rows) x 40 ft long. Corn planting dates ranged from early May to late May depending on year and location of the trial. Targeted population was 34,000 seeds/ac and the hybrid used were Weaver W2021 (2023 and 2024) and Ag Alumni AP4003, ConAgra Vogel, and Crookham Robust (2025 and 2026). Nitrogen fertilizer was applied either as 28% UAN or 32% UAN and was applied as 40 lbs N/ac in a 2x2 starter + remaining required N based on total N rate treatment coulters-injected as a sidedress application at all locations at ~V5 growth stage. Grain yield will be harvested from the center 2 rows at each plot using a Wintersteiger plot combine equipped with a HarvestMaster weigh system. Plant samples were taken at the V8, R1, and R6 growth stages to track nutrient uptake differences between the popcorn hybrids. In addition, grain samples will be taken prior to harvest for quality, popping, and kernel specific data.

On-Farm Locations:

Research trial treatment designed included randomized completed block designs with 4 - 5 replications. Plot size measured 30 - 40 ft wide (12 - 16, 30-inch corn rows, depending on farm planter size) x the length of the field. Targeted population and variety were chosen by the participating farmer. Nitrogen fertilizer was applied either as 28% UAN or 32% UAN at all locations and was coulters-injected sidedress at all locations at ~V5 growth stage. Total sidedress rates applied were adjusted depending on the presence of starter N used. Grain yield was harvested from the center 8 - 12 rows using commercial combines with calibrated yield monitors.



Figure 1. Lower canopy photos highlighting nitrogen deficiency (leaf firing) differences between different N rate treatments in West Lafayette, IN. From left to right the N rates are 0 lbs N/acre, 120 lbs N/acre, and 240 lbs N/acre.



Figure 2. Ear development differences sampled from research plots in West Lafayette, IN on August 19, 2024. From left to right, individual ears were sampled from plots with 240, 180, 120, 60, and 0 lbs N/acre.



Figure 3. Aerial photo of the popcorn hybrid x nitrogen fertilizer rate trial in West Lafayette, IN. 2025.

Research Results – Grain Yield Response to Nitrogen Rate:

The objective of this research trial was to first quantify popcorn response to nitrogen fertilizer rate in comparison to dent corn (2023 and 2024) and to compare popcorn variety response to nitrogen fertilizer rate (2025 and 2026). Throughout these research trials we set up both research station and on-farm research locations with multiple nitrogen fertilizer rates. The goal of this research is to assess popcorn yield, plant, and quality response to different nitrogen rates and also determine the exact nitrogen fertilizer rate where yield is maximized in popcorn or the Agronomic Optimum Nitrogen Rate (AONR). Figure 4 below shows an example of how this data from the field trials looks following harvest and how we utilize regression analysis to examine the nitrogen rate response and determine the nitrogen rate required at each location and hybrid type to maximize yield (e.g., AONR).

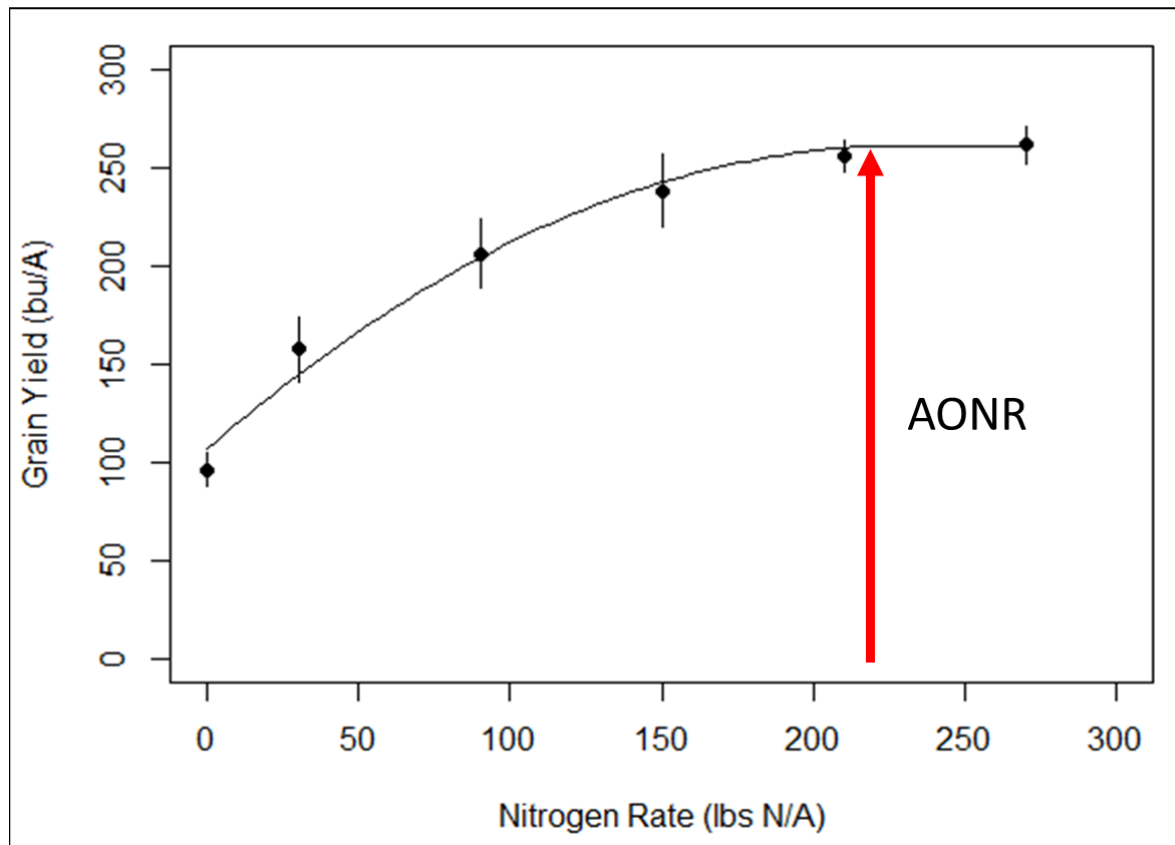


Figure 4. Example of Agronomic Optimum Nitrogen Rate (AONR) calculation using quadratic-plateau regression. The AONR is determined as the exact nitrogen fertilizer rate (lbs N/ac) required to maximize corn yield.

Table 1. Quadratic plateau regression analysis parameters and calculated agronomic optimum nitrogen fertilizer rates (AONR) and yields obtained at the calculated AONR values for popcorn research trials. West Lafayette, Delphi, and Oaktown, IN. 2023.

Location	Regression Parameters			AONR	YAONR†	
	Model†	Intercept	Linear Coefficient			
			Quadratic Coefficient	lbs N ac ⁻¹	lbs ac ⁻¹	
West Lafayette	QP	3804.3	53.97	-0.203	132.67	7384.34
Delphi	QP	5486.2	14.65	-0.045	162.34	6675.63
Oaktown	QP	2861.3	18.79	-0.069	134.43	4125.01

† QP, quadratic-plateau regression model; YAONR, popcorn grain yield at the agronomic optimum N rate (AONR).

Table 2. Quadratic plateau regression analysis parameters and calculated agronomic optimum nitrogen fertilizer rates (AONR) and yields obtained at the calculated AONR values for *popcorn* research trials. West Lafayette, Wanatah, and Butlerville, IN. 2024.

Location	Regression Parameters			AONR	YAONR†	
	Model†	Intercept	Linear Coefficient			Quadratic Coefficient
					lbs N ac ⁻¹	lbs ac ⁻¹
West Lafayette (On-Farm)	QP	4614.1	29.33	-0.119	133.1	6419.7
West Lafayette (ACRE)	QP	5035.9	17.17	-0.058	146.1	6289.6
Wanatah	QP	4373.2	19.63	-0.054	179.7	6137.7
Butlerville	QP	3602.4	26.11	-0.077	168.3	5800.1

† QP, quadratic-plateau regression model; YAONR, popcorn grain yield at the agronomic optimum N rate (AONR).

Table 3. Agronomic Optimum Nitrogen Rate (AONR) comparisons between popcorn and dent corn hybrids. Data was combined across all research trial site-years to provide an estimate of how much less nitrogen fertilizer is required for popcorn in comparison to dent corn.

Corn Type	Site-Year Average AONR (2023-25)	Site-Year Average YAONR (2023-25)
	---- lbs N/ac ----	----- lbs/ac ----
Popcorn	157.4	6013
Dent Corn	200.6	15161
Difference (Popcorn vs Dent)	-43.2	-

Across all research trials in 2023 and 2024, the average agronomic optimum nitrogen rate (AONR) required for popcorn was 157 lbs N/ac whereas, for dent corn the average AONR was 200 lbs N/ac (Table 3). **This data shows that popcorn requires ~40 to 50 lbs N/ac less than what is required by dent corn when compared side by side in the same environment.**

Table 4. Quadratic plateau regression analysis parameters and calculated agronomic optimum nitrogen fertilizer rates (AONR) and yields obtained at the calculated AONR values for the different hybrids on the popcorn research trials. West Lafayette (ACRE), Wanatah (PPAC), and Butlerville (SEPAC), IN. 2025.

Location	Hybrid	Model [†]	Intercept	Linear Coefficient	Quadratic Coefficient	R ²	AONR [†]	YAONR [†]
							-lbs N/ac-	--lbs/ac--
ACRE	Conagra	QP	67.82	0.34	-0.0012	0.67	135.04	5,238
	Crookham	QP	51.68	0.41	-0.0013	0.62	158.02	6,125
	Ag Alumni	QP	72.00	0.16	-0.0004	0.16	198.01	6,968
PPAC	Conagra	QP	59.79	0.30	-0.0009	0.45	159.18	4,981
	Crookham	QP	62.97	0.38	-0.0013	0.56	145.20	6,580
	Ag Alumni	QP	58.14	0.54	-0.0021	0.91	130.25	7,960
SEPAC	Conagra	QP	35.13	0.47	-0.0013	0.66	181.98	2,885
	Crookham	QP	46.79	0.37	-0.0009	0.66	204.43	4,756
	Ag Alumni	QP	47.81	0.44	-0.0011	0.69	196.28	5,839

[†] QP, quadratic-plateau regression model; YAONR, popcorn grain yield at the agronomic optimum N rate (AONR).

2025 research trial data compared the grain yield response to nitrogen rate across different popcorn varieties (Table 4). In 2025, popcorn varieties did differ in their yield response to nitrogen fertilizer rate and with responses differing based on location. On average, the Ag Alumni and Crookham varieties required higher agronomic optimum nitrogen rates (AONR) in comparison to the Conagra variety which is likely a reflection of popcorn variety yield potential. However, these research trials are being repeated in 2026 to help confirm this result.

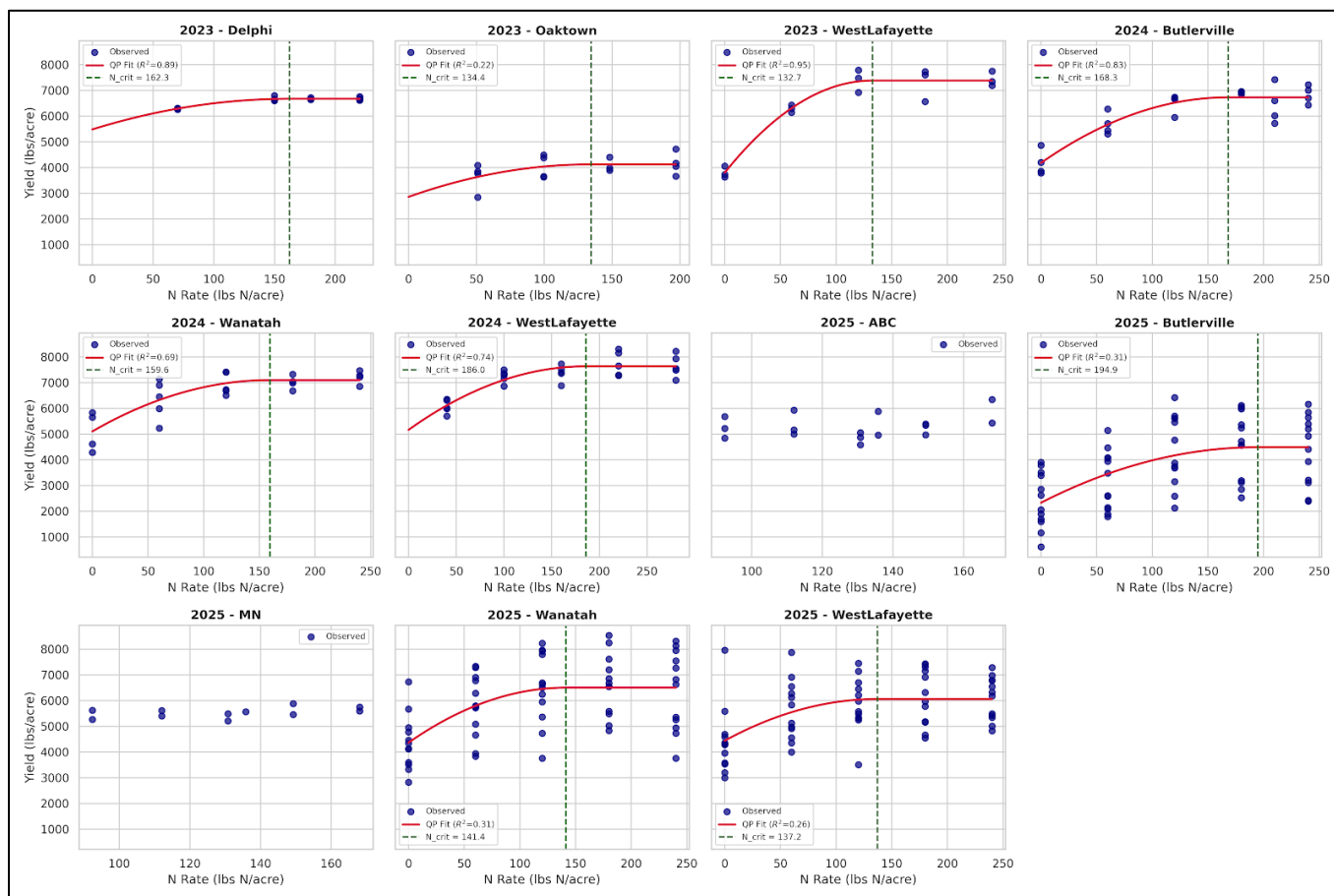


Figure 5. Individual figures which show the individual data points and the popcorn yield response to nitrogen fertilizer rate across all Indiana research locations (2023 – 25).

Research Results – Plant Nitrogen Uptake Data:

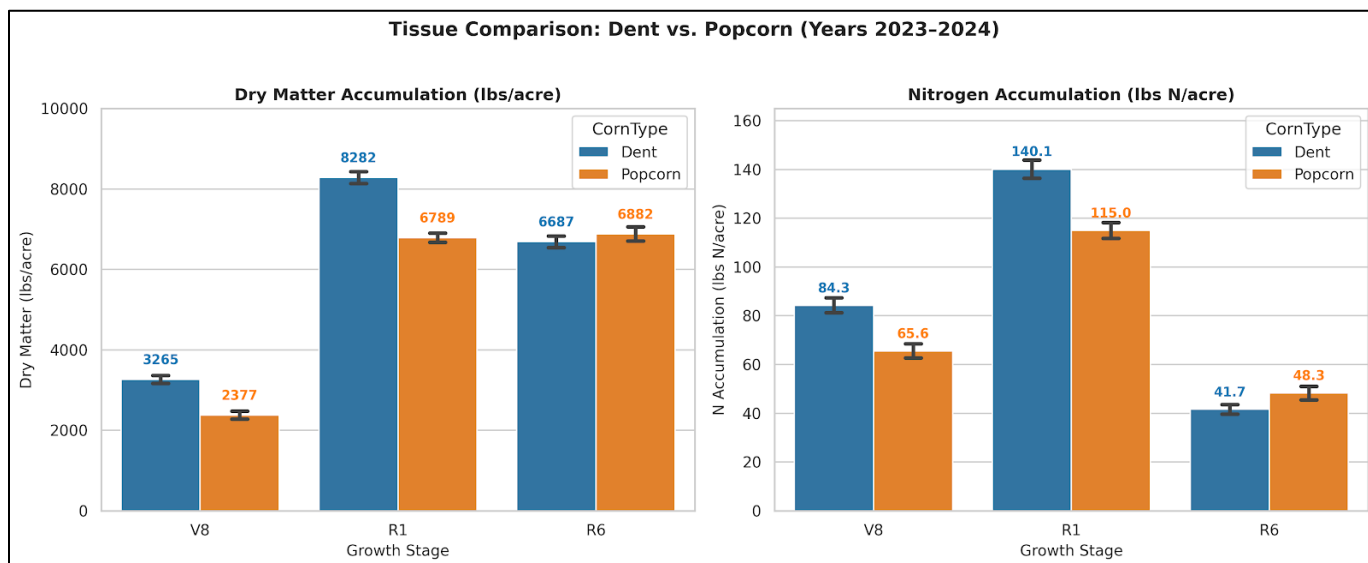


Figure 6. Whole plant dry matter and whole plant nitrogen uptake differences between dent corn (Pioneer 1108Q) and popcorn (Weaver W2021) at growth stages V8 (8 collared leaves), R1 (silking), and R6 (physiological maturity). Data combined across years and locations. Wanatah, West Lafayette, and Butlerville, IN 2023-24

Field evaluations comparing popcorn and dent corn revealed distinct differences in dry matter accumulation and nitrogen uptake throughout the growing season (Figure 6). During the early vegetative stage (V8), dent corn exhibited substantially greater vigor, producing 37.4% more dry matter (888 lb/ac) and accumulating 28.5% more nitrogen in the plant (18.7 lb N/ac) than popcorn (Figure 6). This greater early-season growth reflects the more rapid leaf area development and biomass accumulation characteristic of dent corn. Both crop types reached their maximum dry matter production and nitrogen uptake at the R1 (silking) stage, emphasizing the importance of ensuring adequate nitrogen availability through pre-plant and sidedress applications prior to pollination. However, dent corn produced 22% more biomass (1,493 lb/ac) and accumulated 21.9% more nitrogen (25.2 lb N/ac) than popcorn. As the crop matured (R6 growth stage), nitrogen and carbohydrates were remobilized from vegetative tissues to the developing grain in both crop types. Popcorn, however, retained slightly more biomass (2.8%) and substantially more nitrogen (13.7%) in the plant at physiological maturity, suggesting reduced nitrogen remobilization to the grain as compared to dent corn. Overall, these findings indicate that dent corn has greater early-season nitrogen demand, higher total whole plant nitrogen uptake, and higher remobilization need of nitrogen from plant tissue to the grain at maturity. These results further reflect why popcorn nitrogen requirements are lower than dent corn when grown in the same environment.

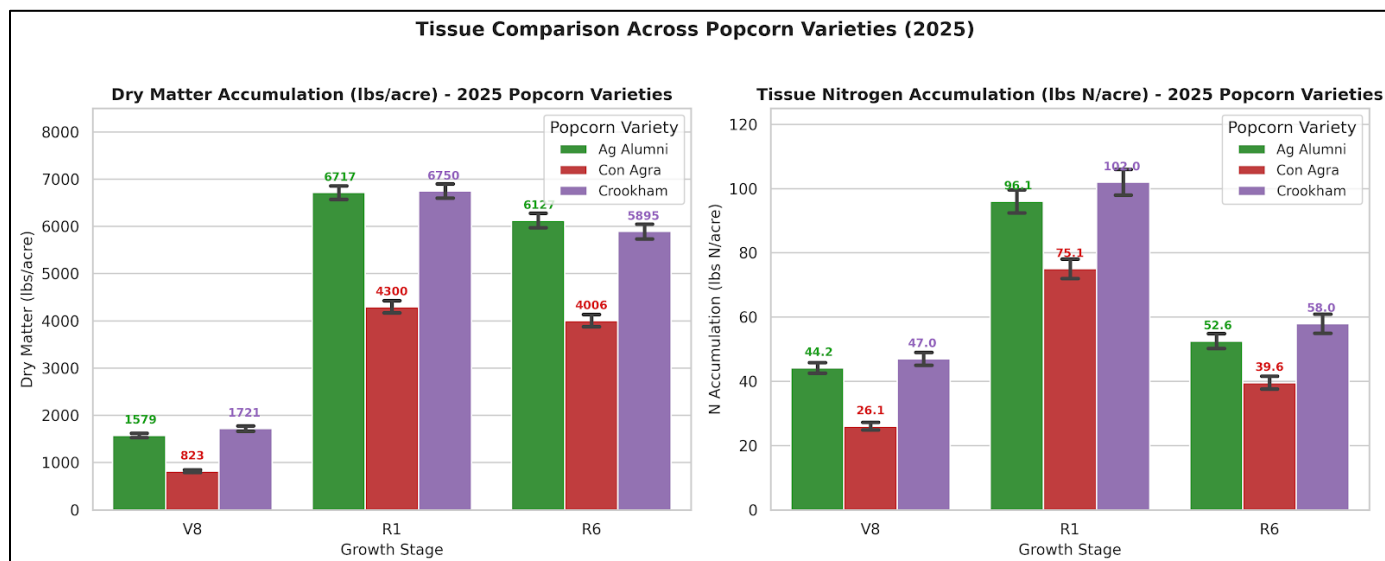


Figure 7. Whole plant dry matter and whole plant nitrogen uptake differences between different popcorn varieties (Ag Alumni, Conagra, and Crookham) at growth stages V8 (8 collared leaves), R1 (silking), and R6 (physiological maturity). Data combined across locations. Wanatah, West Lafayette, and Butlerville, IN 2025.

Table 4. Whole plant dry matter and whole plant nitrogen uptake differences between different popcorn varieties (Ag Alumni, Con Agra, and Crookham) at growth stages V8 (8 collared leaves), R1 (silking), and R6 (physiological maturity). Data combined across locations. Wanatah, West Lafayette, and Butlerville, IN 2025.

Corn Growth Stage	Popcorn Variety	Mean Whole Plant Dry Matter --- lbs/ac ---	Mean Whole Plant Nitrogen ---- lbs N/ac ---
V8 (8 collared leaves)	Ag Alumni	1579	44.2
	Conagra	823	24.9
	Crookham	1721	48.4
R1 (silking)	Ag Alumni	6717	98.1
	Conagra	4300	75.9
	Crookham	6750	107.8
R6 (maturity)	Ag Alumni	6127	55.3
	Conagra	4006	37.1
	Crookham	5895	60.0

The 2025 research trial comparing whole-plant biomass production and nitrogen uptake among popcorn varieties demonstrated significant differences in growth and nitrogen accumulation throughout the growing season (Figure 7 and Table 4). Across all growth stages, Crookham consistently produced the greatest whole-plant dry matter and accumulated the most nitrogen, followed closely by Ag Alumni, while Conagra exhibited substantially lower biomass production and nitrogen uptake. At the V8 stage,

Crookham and Ag Alumni produced approximately twice the biomass and nitrogen uptake of Conagra, indicating more vigorous early-season growth. These differences persisted through the R1 (silking) stage, where maximum biomass and nitrogen accumulation occurred, with Crookham reaching 6,750 lb/ac of dry matter and 107.8 lb N/ac compared with only 4,300 lb/ac and 75.9 lb N/ac for Conagra. As expected, whole-plant biomass and nitrogen content declined by physiological maturity (R6) as nutrients were remobilized from vegetative tissues to the developing grain. However, Crookham maintained the greatest residual biomass and nitrogen, followed by Ag Alumni, while Conagra remained the lowest. These findings demonstrate that popcorn variety influences both biomass production and nitrogen uptake dynamics throughout the season, highlighting the importance of considering varietal differences when developing nitrogen management strategies. In addition, this data provides additional insight in that different popcorn varieties may require different optimum nitrogen rates which is likely dictated by variety vigor, biomass production, and yield potential. This popcorn variety research trial is being repeated in 2026 which will allow for improved understanding of trial results.

Research Results – Popping Data Influenced by Nitrogen Fertilizer:

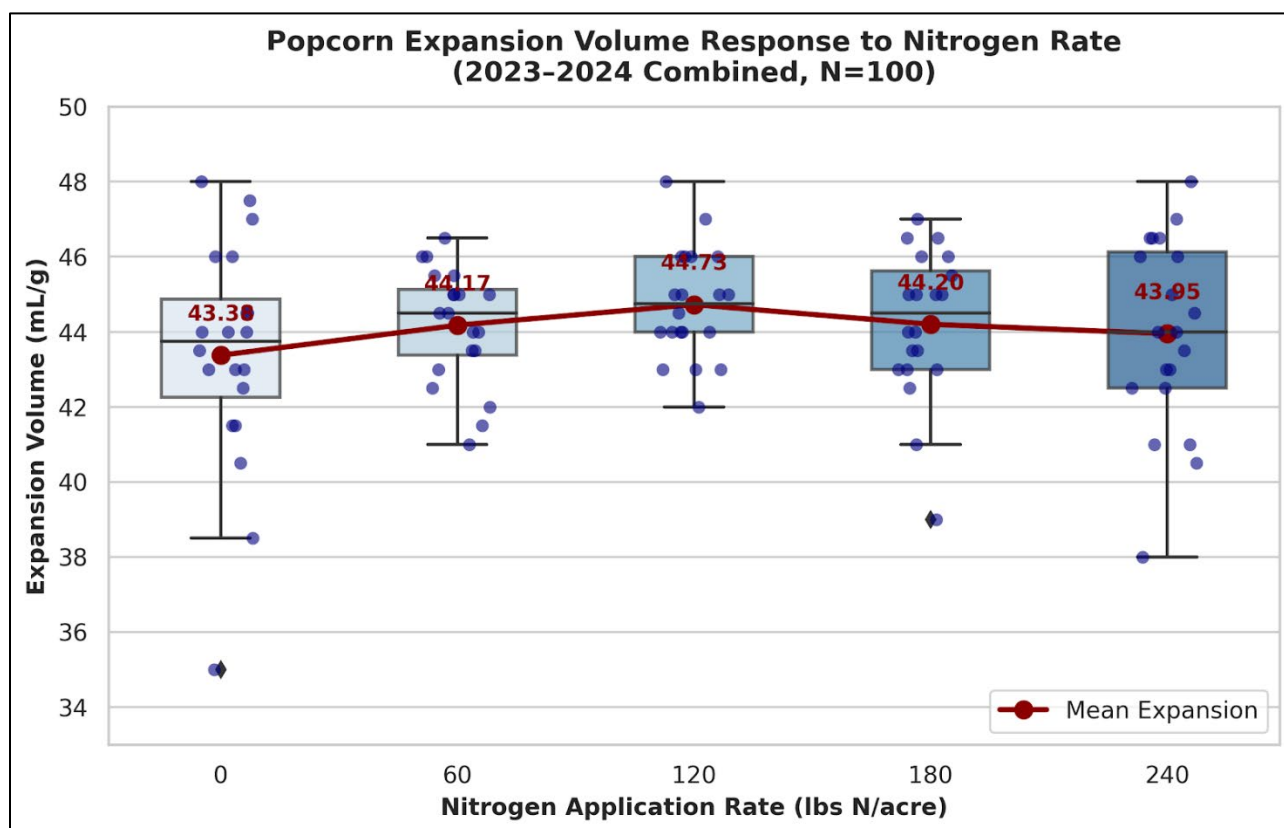


Figure 8. Popcorn variety (Weaver W2021) popping expansion response to nitrogen fertilizer rate. Data combined across locations and years. Wanatah, West Lafayette, and Butlerville, IN 2023-24.

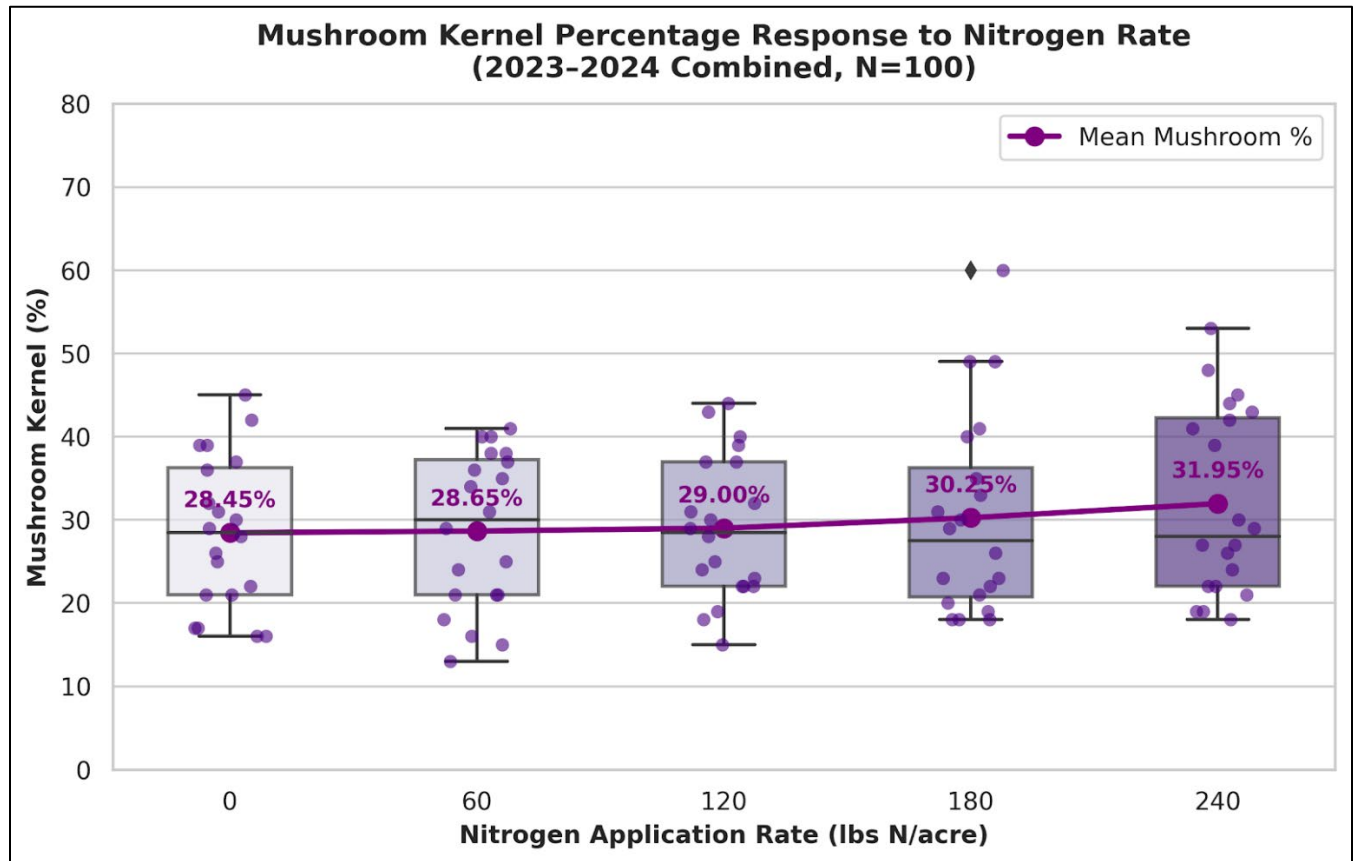


Figure 9. Popcorn variety (Weaver W2021) popping mushroom kernel percentage in response to nitrogen fertilizer rate. Data combined across locations and years. Wanatah, West Lafayette, and Butlerville, IN 2023-24.

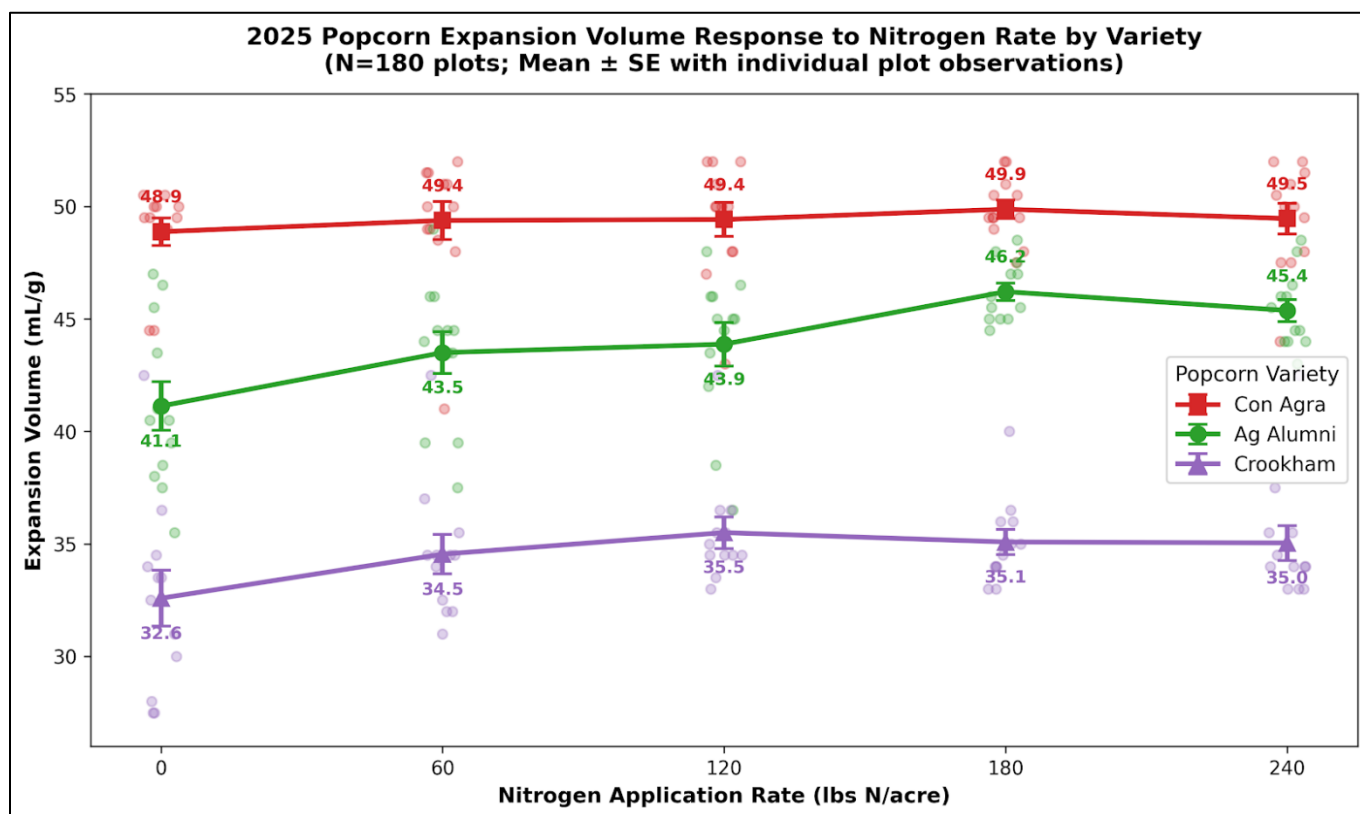


Figure 10. Popcorn variety type (Ag Alumni, Conagra, and Crookham) popping expansion in response to nitrogen fertilizer rate. Data combined across locations. Wanatah, West Lafayette, and Butlerville, IN 2025.

Table 5. Popcorn hybrid mushroom %, butterfly %, kernel size, and expansion in response to applied nitrogen (N) fertilizer rates. Data combined across all locations in the research trial. West Lafayette, Wanatah, and Butlerville, IN (2025).

Hybrid	N Rate	Mushroom	Butterfly	Kernel Size	Expansion
	--- lbs N/ac ---	--- % ---	--- % ---	--- kernels/10 g ---	--- mL/g ---
Conagra	0	24.9 a*	75.1 a*	63.6 a*	48.8 a*
	60	22.8 a	77.2 a	59.1 a	49.4 a
	120	26.4 a	73.6 a	60.5 a	49.4 a
	180	20.8 a	79.2 a	61.3 a	49.9 a
	240	25.5 a	74.5 a	61.4 a	49.5 a
Crookham	0	87.8 a	12.2 a	51.5 a	32.6 a
	60	88.0 a	12.0 a	48.5 ab	35.0 a
	120	89.7 a	10.3 a	45.7 b	34.9 a
	180	89.6 a	10.4 a	45.7 b	34.6 a
	240	89.4 a	10.6 a	46.1 ab	35.0 a
Ag Alumni	0	24.3 a	75.7 a	64.1 a	41.1 b
	60	25.2 a	74.8 a	61.9 a	43.7 ab
	120	25.6 a	74.4 a	62.8 a	44.0 ab
	180	24.9 a	75.1 a	59.8 a	46.2 a
	240	18.0 a	82.0 a	65.3 a	45.4 a

*Mean values within each individual parameter/column that do not contain the same letter are statistically different ($P < 0.1$).

Conclusions:

- Popcorn required ~40 to 50 lbs N/ac less than dent corn for maximum grain yield when grown in the same environment.
- Popcorn has ~22% less total whole plant dry matter production and also ~22% less total whole plant nitrogen uptake at the R1 growth stage (silking) as compared to dent corn which further reflects the lower N rate requirement.
- Nitrogen fertilizer rate may impact both kernel expansion and the ratio of percent kernel mushroom to butterfly, however this response seems to be variety dependent. Overall, peak expansion is achieved at the optimum nitrogen rate required. In addition, in some cases, too much nitrogen fertilizer was detrimental to overall expansion.
- Lower yield popcorn varieties may require less total nitrogen fertilizer than higher yield varieties. However, additional data is being collected in 2026 to confirm this result.